

Project 17

Global IK-Map Planning for KUKA iiwa 7

Format: Group of two students; simulation-based project.

Prerequisites: Redundant inverse kinematics, geometric Jacobians, joint-limit filtering, and graph-based path planning.

Motivation

Choosing a redundant angle independently at every pose can produce apparently feasible configurations that do not connect into a trajectory. A global view of path progress and redundancy exposes which IK families survive and whether a constant posture parameter must be replaced by a varying schedule.

Task

Overall objectives. Build classified IK feasibility charts for one iiwa 7 full-pose path, then compare constant- q_3 choices with a continuous varying- q_3 schedule through validated feasible connections.

- **Validate the path and model.** Inspect the URDF, D–H conventions, physical limits, and tool frame. Read the supplied pose CSV and check quaternion normalization, component order, and pose continuity. Reuse the supplied analytical IK and verify its outputs by full forward kinematics.
- **Enumerate and classify the chart.** Let $\lambda = q_3$ and evaluate each pose sample over a sorted redundancy grid. Retain the supplied analytical labels for up to eight fixed-angle candidates, including missing-solution slots. Explain that labels in a fixed-angle slice are not by themselves proof of global component membership.
- **Apply explicit feasibility checks.** Record position/orientation residuals, joint-limit margins, full-Jacobian regularity, and reduced-chart regularity. Produce separate path-sample versus redundant-angle panels for each label with failure reasons. Plot the smallest singular value of the scaled 6×7 Jacobian separately from the square reduced determinant.
- **Compare constant and varying schedules.** Test each sampled constant- λ row for complete-path continuation. Build a layered graph or dynamic program over neighboring feasible candidates to minimize joint travel. An edge is accepted only after intermediate poses are re-solved and checked; proximity in the image is not sufficient.
- **Validate the selected trajectory.** Re-solve the final schedule on a denser path grid and compare its completion, travel, and minimum margins with the best constant-angle candidate. Preserve actual joint continuity within physical limits. Report any topology or connection that remains unresolved at the chosen resolution.

Specifics and scope

Use one supplied pose path and enumerate the fixed- q_3 IK candidates on a reference grid of 175 negative and 175 positive redundancy angles, excluding zero, with bounds documented from the supplied robot model. For development, a smaller grid is acceptable; perform the final comparison on the reference grid or justify an adaptive equivalent. Limit refinement to one critical corridor. Planning across chart folds is outside this project’s core scope.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand the iiwa model, fixed-angle IK classification, and how feasibility varies over path progress and redundancy. Deliver branch-labelled charts, a validated schedule where found, and a quantitative explanation of the difference between constant and varying redundancy.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 robot description (URDF) and visualization meshes (STL) will be provided to students, together with a mathematical continuation model with unrestricted joints and an analytical fixed- q_3 IK model. Reuse the supplied IK rather than deriving a general 7R solver. Apply the physical joint limits from the robot description separately when using the unrestricted model. Check D–H parameters, joint axes, and the fixed tool transform against the URDF, and validate IK outputs by independent forward kinematics.

A sampled rectangular full-pose path and a reference fixed- q_3 IK table will also be provided. The pose data contain position coordinates and quaternion columns $\mathbf{q}_w, \mathbf{q}_x, \mathbf{q}_y, \mathbf{q}_z$; normalize the quaternions and make their signs consistent before interpolation. The reference IK table checks solutions at $q_3 = 30^\circ$ only; it does not replace evaluation at other angles. Use its configurations to check pose residuals and branch labels, applying physical joint limits separately.

Implement the redundancy-grid enumeration, connection checks, and schedule selection using the supplied IK. Build a layered graph over the resulting candidates, validate edges at intermediate poses, and visualize the selected trajectory with the supplied meshes.